

Algorithm and Dataset

Project Title

Loan Approval Prediction System Using Machine Learning

1. Algorithm

Introduction

The Loan Approval Prediction System is developed using **Machine Learning** to automate the loan approval process. The system analyzes historical loan application data and predicts whether a new loan application should be approved or rejected. It follows a supervised learning approach because the model is trained using labeled data, where the loan status is already known. The proposed system uses the **Random Forest Classifier**, a powerful ensemble machine learning algorithm that combines multiple decision trees to produce accurate and reliable predictions. Random Forest is chosen because it provides high accuracy, reduces overfitting, and handles both numerical and categorical data efficiently.

Step-by-Step Procedure

Step 1: Data Collection

- Collect the loan approval dataset from a reliable source such as Kaggle.
- The dataset contains previous loan application records along with their approval status.

Step 2: Import Required Libraries

- Import Python libraries including Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn.

Step 3: Load the Dataset

- Read the CSV file using Pandas.
- Display the dataset and check the number of rows, columns, and data types.

Step 4: Data Preprocessing

- Handle missing values.
- Remove duplicate records.
- Convert categorical data into numerical values using encoding techniques.
- Perform feature scaling if necessary.

Step 5: Exploratory Data Analysis (EDA)

- Analyze the distribution of applicant features.
- Visualize relationships among different variables.
- Identify the most influential features affecting loan approval.

Step 6: Feature Selection

- Separate the independent variables (features) from the dependent variable (Loan Status).
- Select relevant attributes that contribute to prediction accuracy.

Step 7: Split the Dataset

- Divide the dataset into training and testing sets.
- Use 80% of the data for training and 20% for testing.

Step 8: Model Training

- Train the Random Forest Classifier using the training dataset.
- The model learns patterns from historical loan records.

Step 9: Model Evaluation

- Test the trained model using the testing dataset.
- Evaluate performance using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.

Step 10: Loan Prediction

- Enter the applicant's details such as income, loan amount, credit history, education, and employment status.
- The trained model predicts whether the loan application should be approved or rejected.

Logic or Method Used

The project follows a **Supervised Machine Learning Classification** approach. Historical loan application records are used to train the model. During training, the algorithm learns the relationship between applicant details and loan approval status. When new applicant information is entered, the trained model compares it with previously learned patterns and predicts whether the applicant is eligible for loan approval.

The prediction is based on several important applicant attributes, including:

- Gender
- Marital Status
- Education
- Self Employment
- Number of Dependents
- Applicant Income
- Co-applicant Income
- Loan Amount
- Loan Amount Term
- Credit History
- Property Area

Among these, **Credit History** is considered one of the most significant factors influencing loan approval.

Model or Technique Applied

The project uses the **Random Forest Classifier**, which is an ensemble learning algorithm based on multiple decision trees.

Machine Learning Technique

- Supervised Learning

Problem Type

- Binary Classification

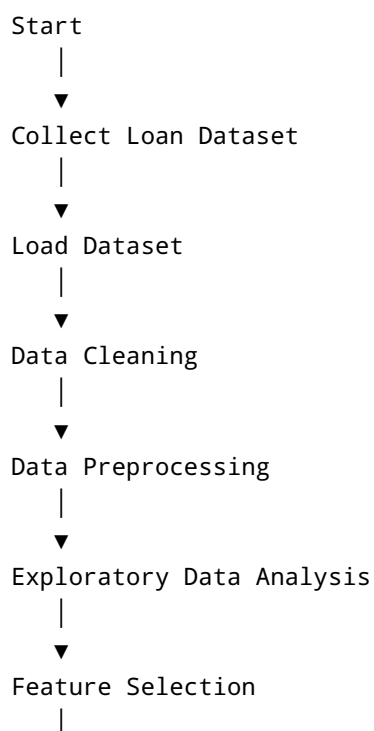
Output Classes

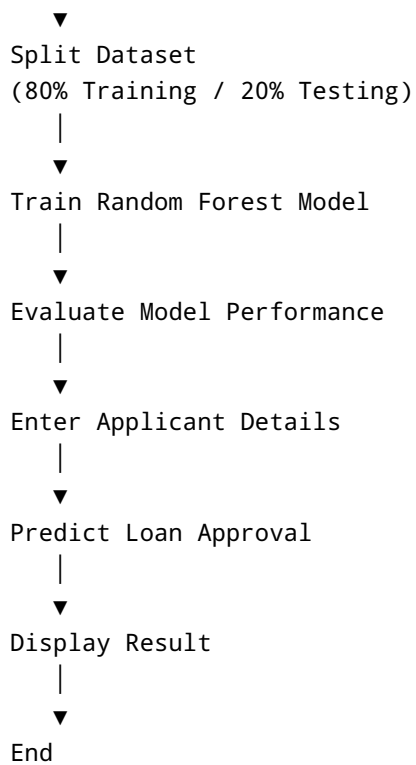
- Loan Approved (Yes)
- Loan Rejected (No)

Advantages of Random Forest

- High prediction accuracy
- Handles large datasets efficiently
- Reduces overfitting
- Works well with categorical and numerical features
- Provides reliable and stable predictions

Process Flow of the System





2. Dataset

Dataset Overview

The Loan Approval Prediction System uses a historical loan application dataset to train and evaluate the machine learning model. The dataset contains applicant information and loan approval status, allowing the model to identify patterns that influence loan approval decisions.

Source of Data

The dataset is obtained from the **Kaggle Loan Prediction Problem Dataset**, which is widely used for machine learning and predictive analytics projects.

Number of Records and Samples

- Total Records (Samples): **614**
- Total Features (Columns): **13**
- Target Variable: **Loan_Status**

Features or Attributes

The dataset contains the following attributes:

Attribute	Description
Loan_ID	Unique loan application ID
Gender	Male or Female
Married	Marital status
Dependents	Number of dependents
Education	Graduate or Not Graduate
Self_Employed	Employment status
ApplicantIncome	Monthly income of applicant
CoapplicantIncome	Monthly income of co-applicant
LoanAmount	Requested loan amount
Loan_Amount_Term	Loan repayment duration
Credit_History	Previous credit history
Property_Area	Urban, Semiurban, or Rural
Loan_Status	Approved (Y) or Rejected (N)

Training and Testing Data Split

The dataset is divided into two parts:

Dataset	Percentage
Training Dataset	80%
Testing Dataset	20%

Approximately:

- Training Records: **491**
- Testing Records: **123**

The training dataset is used to build the prediction model, while the testing dataset is used to evaluate its performance on unseen data.

Data Format and Type

File Format

- CSV (Comma Separated Values)

Data Types

Numerical Attributes

- ApplicantIncome
- CoapplicantIncome
- LoanAmount
- Loan_Amount_Term

Categorical Attributes

- Gender
- Married
- Dependents
- Education
- Self_Employed
- Credit_History
- Property_Area
- Loan_Status

Before training the model, categorical attributes are converted into numerical form using encoding techniques so that they can be processed by the machine learning algorithm.

Conclusion

The Loan Approval Prediction System uses the **Random Forest Classifier** to predict loan approval based on historical applicant data. The project follows a structured workflow consisting of data collection, preprocessing, exploratory data analysis, feature selection, model training, evaluation, and prediction. The dataset contains **614 loan application records** with **13 important features**, making it suitable for supervised machine learning classification. By automating the loan approval process, the system reduces manual effort, improves prediction accuracy, minimizes processing time, and supports financial institutions in making faster and more reliable lending decisions.